Agentic Workflow: Intelligent ERPNext Implementation System

Purpose

This document describes the agentic workflow used to migrate an existing company's data (Excel, PDF, CSV, or any file format) into ERPNext in a **secure, structured, and intelligent way**.

The goal is not just data import, but **full ERP implementation with company context**, industry alignment, and validation at every step.

High-Level Objective

- Accept raw company data in any format
- Secure and sanitize all inputs
- Understand the company as a business (not just rows and columns)
- Decide the correct ERP structure for that company
- Map and validate real data against ERPNext schemas
- Import safely with user confirmation

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1. Ingestion & Storage (Quarantine Stage)

What happens

- The user uploads files (Excel, PDF, CSV, ZIP, etc.).
- Files are treated as **untrusted** at this stage.

Storage

- All uploaded files are stored in:
- **MinIO – Quarantine Bucket**

Rules

- No parsing or AI processing happens here.
- Files are immutable.
- Only the virus scanning service has access.

Metadata Stored

- Client ID
- Upload timestamp

- File type
- Checksum / hash

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2. Virus Scanning Service

Purpose

Ensure no malicious files enter the system.

Flow

- Files are scanned for:
- Viruses
- Malware
- Suspicious payloads

Outcome

- **Clean files** → moved to `clean_data` bucket in MinIO
- **Infected files** → rejected and user notified

Output

- Scan result stored in database
- Scan confidence recorded

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3. Company Profile Generation Service (Core Intelligence)

This service builds **business understanding**, not ERP data.

Inputs

- Clean files from MinIO
- User-provided company details (if any)
- Industry / niche knowledge base
- Previous implementation cases

3.1 Entity & Signal Extraction

The service extracts high-level business signals such as: - Company name - Industry and niche - Products and services - Customers and suppliers - Locations - Scale (employees, volume, turnover) - Accounting style - Manufacturing / trading indicators - Compliance hints

This data is used only for **context and decision-making**, not direct ERP import.

3.2 Two Types of Company Profile

A. Absolute Profile (Stable)

Represents the **fundamental nature of the company**.

Changes only when core business characteristics change, such as: - Industry change - Manufacturing to trading (or vice versa) - Major scale shift - Compliance regime change

- Versioned
- Rarely updated

B. Context Profile (Dynamic)

Represents the **current state of interaction** with the system.

Updated on every: - File upload - User correction - Answer to system questions - Mapping or validation change

- Continuously versioned
- Feeds all downstream agents
- Keeps the system adaptive and intelligent

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4. Target Structure Resolution (Industry → ERPNext)

Problem Addressed

ERPNext is flexible, but every industry requires a **different configuration**.

What This Step Does

Based on the company profile, the system decides: - Required ERP modules - Required doctypes - Mandatory fields - Workflow and compliance needs

Inputs

- Absolute profile
- Industry and niche
- Company size
- Location and regulatory context
- Past successful implementations

Gap Analysis

The system compares:

- **Client-provided data** vs
- **Target ERPNext structure for that industry**

If required data is missing: - The user is asked **specific, targeted questions** - No generic forms - Only what is strictly required

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5. Data Schema Mapping (Actual ERP Data)

This stage handles **real ERP data**, not profiles.

Data Types Mapped

- Items
- Customers
- Suppliers
- Chart of Accounts
- Opening stock
- Price lists
- BOMs
- Transactions (if applicable)

Mapping Logic

- Each dataset is mapped to:
- ERPNext doctype
- ERPNext field-level schema (JSON)

Example: - Excel column → ERP field

Mappings are: - Explicit - Stored - Versioned - Reusable

No silent assumptions are made.

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6. Validation & Confidence Scoring

Every stage produces a **confidence score**.

Validation Includes

- Schema correctness
- Data type safety
- Mandatory field checks
- Cross-file consistency
- Business logic validation

User Confirmation Gate

Before importing into ERPNext, the user is shown: - What will be created - What will be modified - Assumptions made by the system - Overall confidence score

The system proceeds **only after explicit user approval**.

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7. Integration & Import into ERPNext

Import Process

- Data imported via ERPNext APIs
- Executed as background jobs
- Idempotent and logged
- Rollback-aware

Post-Import Actions

- Success and failure reports generated
- Errors mapped back to source files
- Context profile updated
- Knowledge base improved for future implementations

Summary Mental Model

1. Secure intake
2. Sanitize inputs
3. Understand the business
4. Decide correct ERP structure
5. Map real data carefully
6. Validate with confidence
7. Import safely

This system represents **automated ERP implementation intelligence**, not just data migration.